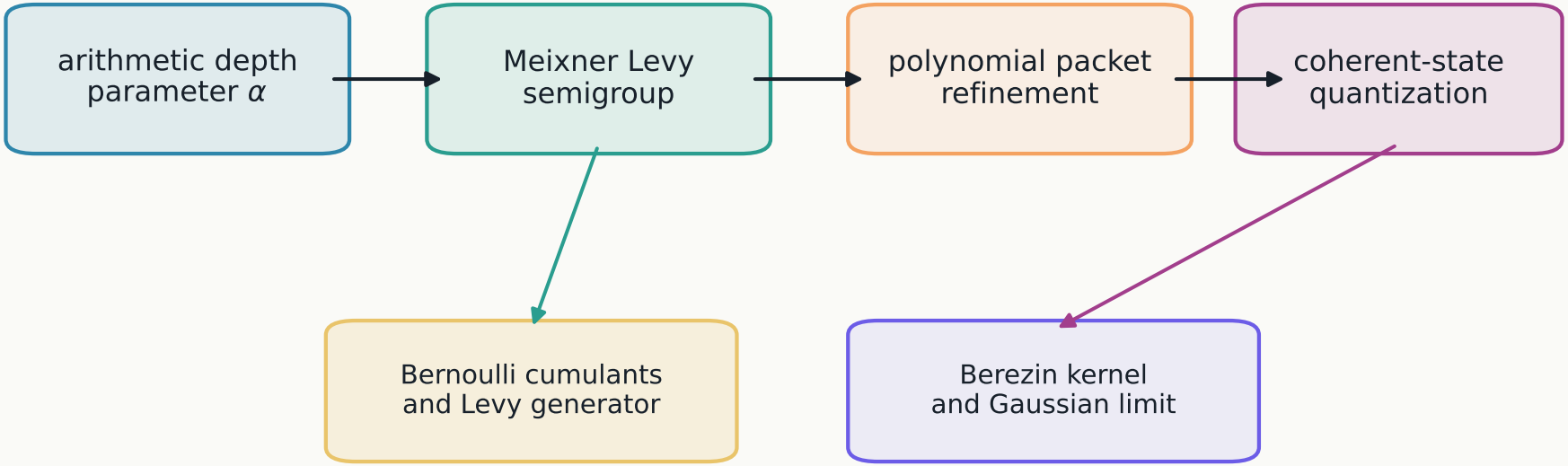Exact transfer hierarchy